

The ARCH fold and the conditional chain

[L3] as a prefix SATZ; [L1] is elementary, not circular

Stefan Hamann

August 30, 2026

Abstract

The r454 limit object is the folded archimedean measure. This note writes the fold weight exactly, identifies the orthogonal polynomials as a Christoffel modification of Chebyshev (not a named Jacobi family), and proves the two remaining lemmas of the prefix program in the only sense the program needs: for each fixed depth n , eventually. [L1] is the three-lag identity plus the support $\Lambda(n) = 0$ for $n < 2$ — no PNT, no zero-free region, no RH. [L3] is $h_j > 0$ for the positive μ -chain and $\rho_n = 0$ on the border-ARCH moment-agreement prefix $J_B \sim \text{BG_DU}/\Delta$. No RH claim. No anti-RH claim.

Status. Ausgang L3_PROVED / L1_PNT_SUFFICIENT / CHAIN_MAPPED. The actual strength of [L1] is *elementary*; PNT is surplus. **Not RH-circular.** No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 The weight

The discrete ARCH fold of a window $(\alpha, M, L = 2M - 2, \Delta = 2\alpha/M)$ is

$$m_j = \sum_{k=1}^{L/2} w_L(\theta_k) T_j(\cos \theta_k), \quad \theta_k = \frac{2\pi k}{L}, \quad (1)$$

$$w_L(\theta) = \frac{2}{L} (1 - \cos \theta) f_A(\theta), \quad (2)$$

$$f_A(\theta) = A(0) + 2 \sum_{p=1}^{M-2} A(p\Delta) \cos(p\theta) + A((M-1)\Delta) \cos((M-1)\theta). \quad (3)$$

$A(s)$ is the tent-smoothed archimedean kernel of the explicit formula (**arch_A_far** / **arch_A_near**): an explicit Gauss–Legendre integral of $e^{-w/2}/(1-e^{-2w})$ against t_Δ . The Nyquist point $k = L/2$ carries the usual half-weight.

The factor $(1 - \cos \theta) = 1 - x$ is a Jacobi $(+\frac{1}{2}, -\frac{1}{2})$ shift of Chebyshev $d\theta$. That candidate is *refuted*: $\int_0^\pi (1 - \cos \theta) \cos(j\theta) d\theta = 0$ for every $j \geq 2$, but $m_2 = 0.1698990367924736 \neq 0$. f_A is a genuine multiplier. For each finite window f_A is a trigonometric polynomial, so w_L is a polynomial times the Chebyshev weight — Christoffel-modified Chebyshev, not a named (α, β) family, and (a_n, b_n) are not closed in A .

f_A changes sign (13 negative DFT modes on kz17, 32 on kz500). The negative fold mass is $2 \cdot 10^{-4}$ (kz17) and $< 10^{-10}$ (kz500). The μ -chain of the program uses only the positive atoms: $h_j > 0$ is the classical fact that a positive discrete measure with N atoms has $h_j > 0$ for $j < N$.

2 The three-lag identity

Lemma 1 (three-lag form). *For the fold (1)–(3),*

$$m_0 = c_0 - c_1, \quad m_j = -\frac{1}{2}c_{j-1} + c_j - \frac{1}{2}c_{j+1} \quad (j \geq 1),$$

to residual $< 10^{-14}$ (kz17 ARCH, $j \leq 8$).

This is discrete orthogonality of $\{1, \cos p\theta\}$ on the circulant grid against $(1 - \cos \theta) \cos(j\theta)$. Every folded moment sees three lags only.

3 $[L1]$ is elementary

Write $c = c^A + c^P$ for the MAIN lags. Linearity of the fold gives $m^{\text{MAIN}} - m^{\text{ARCH}}$ as the three-lag form in c^P . The lag c_i^P is the tent of width Δ centred at $i\Delta$, applied to $-\Lambda(n)n^{-1/2}$. Its support starts at the smallest $u = \log n$ with $\Lambda(n) \neq 0$, i.e. $n = 2$.

Lemma 2 ($[L1]$, elementary). $c_i^P = 0$ whenever $(i+1)\Delta < \log 2$. Hence

$$m_j^{\text{MAIN}} = m_j^{\text{ARCH}} \quad \text{for all } j \leq J_P - 2, \quad J_P = \lceil \log 2 / \Delta - 1 \rceil.$$

On the windows $\{17, 116, 136, 197, 230, 500\}$ one has $n_{\text{stab}} = J_P - 1$ exactly. Sister / oversampled windows have a neighbour $n_{\text{stab}} < J_P$ (conservative). In particular: for each fixed n , every window with $\Delta < \log 2 / (n+2)$ satisfies $m_j^{\text{MAIN}} = m_j^{\text{ARCH}}$ for all $j \leq n$.

Circularity audit. The only arithmetic input is $\Lambda(n) = 0$ for $n < 2$ (definition of Λ) plus the fold identity. PNT is not used. A zero-free region is not used. RH is not used. The identity is stronger than PNT: $|m^{\text{PNT}} - m^{\text{ARCH}}|[2]$ is $6 \cdot 10^{-6}$ (kz17) down to 10^{-10} , while $|m^{\text{MAIN}} - m^{\text{ARCH}}|[2]$ is 10^{-16} . The r452 neighbour rate $a^{-1.9}$ is PNT-carried (MAIN and PNT-world neighbour exponents 2.94 and 2.95) and is *not* load-bearing for eventually. Verdict `L1_PNT_SUFFICIENT` (surplus strength: elementary). **Not** `RH_CIRCULAR`.

The error formula, for the record:

$$m_j^{\text{MAIN}} - m_j^{\text{ARCH}} = \left(-\frac{1}{2}c_{j-1}^P + c_j^P - \frac{1}{2}c_{j+1}^P\right),$$

$$c_i^P = -\sum_n \frac{\Lambda(n)}{\sqrt{n}} t_\Delta(i\Delta - \log n) + (\text{reflection if } \log n < \Delta).$$

PNT replaces the sum by $\int e^{u/2} t_\Delta(i\Delta - u) du$; that is a different classical object, not the intermediary of $[L1]$.

4 $[L3]$ as a prefix SATZ

$\rho_n = f_{b,n}^2 / \eta_n$ with $f_{b,n} = \int p_n d(\text{border})$ and $\eta_n = \int p_n^2 d\sigma$ (r243 / r453).

Lemma 3 (identical measures). *If the border signed measure equals $d\sigma$, then $f_{b,0} = \int d\sigma$ and $f_{b,n} = 0$ for every $n \geq 1$ (orthogonality of the OP system). Verified in exact rationals on a four-atom toy ($\rho_0 = 1$, $f_{b,1} = 0$, $\rho_1 = 0$).*

Lemma 4 (shared moments). *If border and ARCH agree as functionals on $\mathbb{P}_J$, then $f_{b,n} = 0$ for $1 \leq n < J$, hence $\rho_n = 0$.*

The sealed border is the same fold applied to ARCH plus `PB.smooth_comb`, a Riemann sum of $2e^{u/2}$ starting at `BG.DU` = 0.01 (not at $u = 0$). The extra lags begin at $p_B \sim \text{BG.DU} / \Delta$.

Lemma 5 ([L3], eventual per n). *On kx69,137,500 the first 10^{-12} -break between border and ARCH moments is at $J_B \in \{9, 11, 21\}$, matching $\lfloor \text{BG_DU}/\Delta - 1 \rfloor$. Thus $\rho_n = 0$ for $1 \leq n < J_B$. The case $n = 0$ is the mass exception $\rho_0 = M_\partial$. The μ -chain satisfies $h_j > 0$ for $j < N_\mu$. For each fixed $n \geq 1$, every window with $\Delta < \text{BG_DU}/(n + 2)$ has h -prefix positivity through n and $\rho_n = 0 < 5/7$.*

There is no single infinite Jacobi chain of a limit measure: m_0 and m_1 grow with the window (r454). The program never needed that object — it needs the property at each fixed n .

5 The conditional map

Link	Status	Nature
r453 pack algebra $q_n = S_{n-1}/B_w$, $q_n < 1 \Leftrightarrow \rho_n < 5/7$	PROVED	finite algebra
[L2] pack Lipschitz in the moments	MEASURED	elementary, $dq/dm_2 \sim 0.09$
[L1] $m_n^{\text{MAIN}} = m_n^{\text{ARCH}}$	PROVED	elementary (3-lag + $n \geq 2$)
[L3] $h > 0$ and $\rho_n = 0$	PROVED	eventual per n (Δ small)
Lean Iff.rfl prefix $\Leftrightarrow$ full mincut	existing identification	not this round
Weil / explicit formula $\Rightarrow$ RH	OPEN	the residual RH problem; <i>not</i>

Together: for each fixed n , every sufficiently fine window has the prefix mincut at n ($n < J_P$ and $\rho_n = 0$). The Lean identification then gives the full mincut at that n . The remaining arrow from the mincut family to RH is the classical Weil programme and is *not* a claim of this note.

6 Kill table

Kill	Status
Symbolic ρ at small j	sympy rationals: $\rho_1 = 0$ on identical measures
Closed form vs numerical ARCH	3-lag residual $< 10^{-14}$
PNT world, same fold	neighbour rate $\approx$ MAIN; PNT is <i>not</i> the kernel
Determinism	two identical smokes
Silent assumptions	$\Lambda(n) = 0$ for $n < 2$; $\text{BG_DU} = 0.01$ is a construction constant, named

Claim boundary. Finite-window identities and an eventual-per- n prefix SATZ. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.